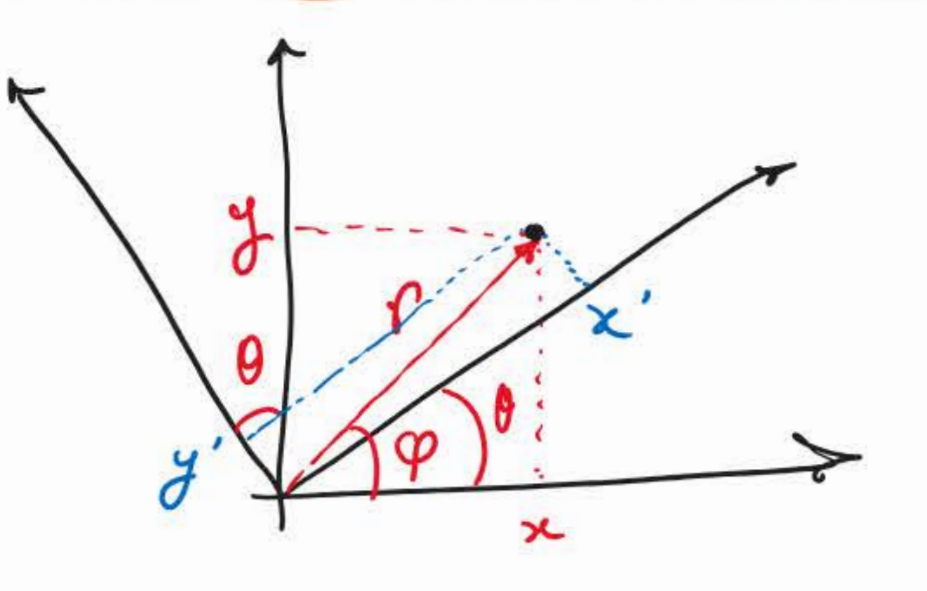
Problem 0. n^3. Yes, matrix multiplication can be done with fewer multiplication operations, by converting them to addition. How much better? Nobody knows. The best result has reduced the number of multiplications to $n^{2.371552}$ as of April 2024

Problem 1.



$x = r\cos\varphi$

$y = r\sin\varphi$ ①

$x' = r\cos(\varphi-\theta)$

$y' = r\sin(\varphi-\theta)$

Recall, $\begin{cases}\cos(\varphi-\theta) = \cos\varphi\cos\theta + \sin\varphi\sin\theta \\ \sin(\varphi-\theta) = \sin\varphi\cos\theta - \cos\varphi\sin\theta\end{cases}$

$$\Rightarrow \begin{cases} x' = \overbrace{r\cos\varphi}^{x}\cos\theta + \overbrace{r\sin\varphi}^{y}\sin\theta = x\cos\theta + y\sin\theta \\ y' = \underbrace{r\sin\varphi}_{y}\cos\theta - \underbrace{r\cos\varphi}_{x}\sin\theta = -x\sin\theta + y\cos\theta \end{cases}$$

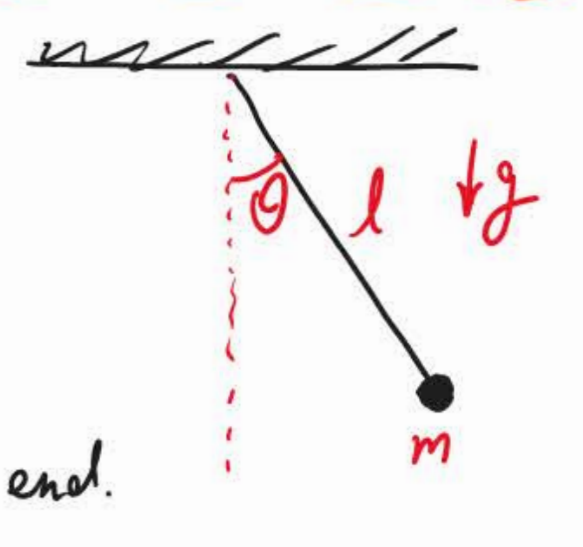
$$\Rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

Problem 2. knowns: g: gravity accl.

θ: angle

l: length of pendulum

m: The pendulum mass at its end.



a) unknown: f: frequency

$$[f] = T^{-1} = [g]^\alpha [l]^\beta [m]^\gamma = \left(\frac{L}{T^2}\right)^\alpha L^\beta M^\gamma$$

$$\Rightarrow \alpha = +\frac{1}{2}, \ \beta = -\frac{1}{2}, \ \gamma = 0$$

$$\Rightarrow \boxed{f \propto \sqrt{\frac{g}{l}}} \text{ or } \boxed{f = f(\theta)\sqrt{\frac{g}{l}}}$$

b) $[\text{Energy}] = [\text{Force}]^\alpha [\text{distance}]^\beta = ML^2T^{-2}$

$= [g]^\alpha [l]^\beta [m]^\gamma = \left(\frac{L}{T^2}\right)^\alpha L^\beta M^\gamma$

$M: \ \gamma = 1$

$L: \ \alpha + \beta = 2$

$T: \ \alpha = 1 \Rightarrow \beta = 1$

$\Rightarrow \boxed{E \propto mgl}$ or $\boxed{E = f(\theta)\, mgl}$

How about $f(\theta)$? E must be rotary and lowest at $\theta = 0$.

$\Rightarrow$ Moreover, the potential energy must change proportional to height:

$h = l(1-\cos\theta)$

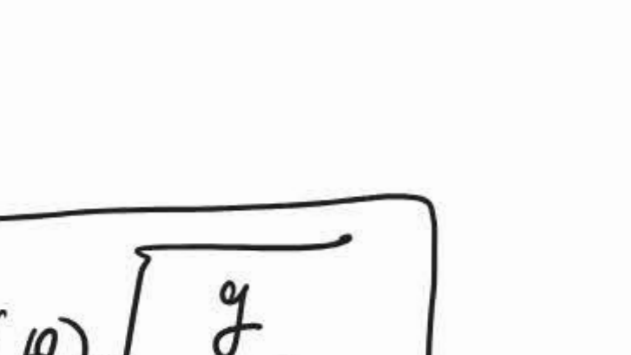
$\Rightarrow \boxed{E = mgl(1-\cos\theta) \text{ likely}}$

Problem 3. known: k: spring constant

m: mass attached to spring

x: displacement from equilibrium

equation: $F = ma = -kx$



a) unknown: $f = ?$ frequency

$$[f] = T^{-1} = [k]^\alpha [m]^\beta [x]^\gamma = \left(\frac{F}{x}\right)^\alpha M^\beta L^\gamma = \left(\frac{M\cancel{L}}{T^2\cancel{L}}\right)^\alpha M^\beta L^\gamma$$

$$\Rightarrow \begin{cases} T^{-1} = T^{-2\alpha} \Rightarrow \alpha = \frac{1}{2} \\ \beta = -1 \\ \gamma = 0 \end{cases}$$

$$\Rightarrow \boxed{f \propto \sqrt{k/m}} \text{ or } \boxed{f = c\sqrt{\frac{k}{M}}}$$

b) $[E] = \frac{ML^2}{T^2} = [k]^\alpha [m]^\beta [x]^\gamma = \left(\frac{M}{T^2}\right)^\alpha M^\beta L^\gamma$

$$\Rightarrow \begin{cases} k: \ \frac{1}{T^2} = \left(\frac{1}{T^2}\right)^\alpha \Rightarrow \alpha = 1 \\ m: \ M = M^\alpha M^\beta = M^{\beta+1} \Rightarrow \beta = 0 \\ x: \ L^2 = L^\gamma \Rightarrow \gamma = 2 \end{cases}$$

$\Rightarrow \boxed{E \propto kx^2}$ or $\boxed{E = \beta kx^2}$

c) knowns: $b, k, m, x, F = ma = -kx + bx^2$

$$[b] = \left[\frac{F}{x^2}\right] = \frac{ML}{T^2L^2} = \frac{M}{T^2L}$$

$$[f] = \frac{1}{T} = [k]^\alpha [m]^\beta [x]^\gamma [b]^\eta = \left(\frac{M}{T^2}\right)^\alpha M^\beta L^\gamma \left(\frac{M}{T^2L}\right)^\eta$$

$$\Rightarrow \begin{cases} M: \ M^0 = M^{\alpha+\beta+\eta} \Rightarrow \alpha+\beta+\eta = 0 \ ① \\ L: \ L^0 = L^{\gamma-\eta} \Rightarrow \gamma - \eta = 0 \ ② \\ T: \ T^{-1} = T^{-2\alpha-2\eta} \Rightarrow 2\alpha + 2\eta = 1 \ ③ \end{cases}$$

①, ③ $\Rightarrow \beta = -\frac{1}{2}$

② $\Rightarrow \eta = \gamma$ ↝ This means x and b must appear together.

$$\rightarrow [f] = [k]^\alpha [m]^\beta [bx]^\gamma = \left(\frac{M}{T^2}\right)^\alpha \left(\frac{M}{T^2}\right)^\gamma \frac{1}{M}$$

$\Rightarrow \boxed{\alpha + \gamma = \frac{1}{2}}$, one possibility: $\alpha = \frac{1}{2}$ as before.

$\Rightarrow \gamma = 0 \Rightarrow \boxed{f = h\left(\frac{xb}{k}\right)\sqrt{\frac{k}{m}}}$

Problem 4) knowns: m: mass of satellite

g: gravity accr.

R: Earth radius

unknowns: $v = ?$

$$[v] = \frac{L}{T} = [m]^\alpha [g]^\beta [R]^\gamma = M^\alpha \left(\frac{L}{T^2}\right)^\beta L^\gamma$$

$$\left.\begin{cases} M: \ \alpha = 0 \\ L: \ L^{\beta+\gamma} = L^1 \\ T: \ T^{-1} = T^{-2\beta} \Rightarrow \beta = \frac{1}{2} \end{cases}\right\} \Rightarrow \gamma = \frac{1}{2}$$

$\Rightarrow \boxed{v \propto \sqrt{gR}}$ or $\boxed{v = c\sqrt{gR}}$

Problem 5) knowns: h: height

$F_d = -m\alpha v$ drag force

Governing equations:

$v(t) = -\frac{g}{\alpha}\left(1 - e^{-\alpha t}\right)$

$y(t) = h - \frac{g}{\alpha}\left(t - \frac{1}{\alpha}\left(1 - e^{-\alpha t}\right)\right)$

unknown: prove the correctness of eqns. for $t \gg \alpha$ or $t \ll \alpha$.

From Taylor expansion:

$$f(x \to a) = f(a) + f'(a)(x-a) + \frac{f''(x)}{2!} + \frac{f^{(3)}(x)}{3!} + \dots$$

$$\Rightarrow e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \text{ Taylor expansion.}$$

If $t \ll \alpha \Rightarrow v(t) = -\frac{g}{\alpha}\left(1 - \exp(-\alpha t)\right) \approx$

$$= -\frac{g}{\alpha}\left(1 - \left[1 - \alpha t + \frac{(\alpha t)^2}{2} - \dots\right]\right)$$

$$\boxed{v(t) = -gt + O\left((\alpha t)^2\right) \text{ for } t \ll \alpha}$$

Does this make sense? Yes, because initially drag force is small for v is small, thus acting like free fall.

$$y(t) = h - \frac{g}{\alpha}\left(t - \frac{1}{\alpha}\left(1 - e^{-\alpha t}\right)\right)$$

$$= h - \frac{g}{\alpha}\left[t - \frac{1}{\alpha}\left[1 - \left(1 - \alpha t + \frac{(\alpha t)^2}{2!} - \dots\right)\right]\right]$$

$$= h - \frac{g}{\alpha^2}\left[\frac{\alpha^2 t^2}{2} + O\left((\alpha t)^3\right)\right] \approx \boxed{h - \frac{gt^2}{2}}$$

Since the contribution of t is zero, we keep the higher term t^2. Again, this makes sense because at the very beginning, the drag force is nearly zero. Therefore, the motion must behave like a free fall.

If $\alpha t \gg 1$. $v(t) = -\frac{g}{\alpha}$ = Constant

This makes sense, because eventually the drag force will become so strong that negate gravity, leading to steady state motion.

$y(t) = -\frac{gt}{\alpha} + h + \frac{g}{\alpha^2}$. This is also consistent with steady motion.

what is the physical meaning of g/α^2.

Problem 6) knowns: m, M masses; $v_M = \frac{(m-M)v}{m+M}$; $v_M = \frac{2mv}{m+M}$

For $m \ll M$:

$v_m = -v$; $v_M = 0$

This makes sense because M is so heavy that it acts like a wall.

$m = M$: $v_m = 0$; $v_M = v$

while not so obvious, this does not seem to contradict our intuition.

$m \gg M$: $v_m = v$; $v_M = +2v$

m is so massive that it acts like a wall. The mass M effectively bounces off this wall after colliding with it with velocity $-v$ with respect to m. Therefore, it must have velocity $2v$ for the stationary observer, after the collision.

Problem 7) known: $\omega = \sqrt{\frac{g}{l\cos\theta}}$; unknown: does this eqn. make sense?

$\theta \to 0$: $\cos\theta \to 1 \Rightarrow \omega = \sqrt{\frac{g}{l}}$ is consistent with problem 2.

$\theta \to 90$: $\cos\theta \to 0 \Rightarrow \omega \to \infty$ Also consistent because pendulum must rotate so fast to defy gravity.

Similarly, $l \to \infty \Rightarrow \omega \to \infty$: It takes more and more time for the pendulum to complete one cycle, and vice versa for $l \to 0$.

Similarly, $g \to \infty \Rightarrow$ gravity is so powerful that pendulum returns to its equilibrium almost instantly $\Rightarrow \omega \to \infty$ ✓

$g = 0 \Rightarrow \omega = 0$: no gravity $\Rightarrow$ no oscillations due to gravity.

Problem 8) known: $a_1 = g\,\frac{3m_2m_3 - m_1(4m_3+m_2)}{m_2m_3 + m_1(4m_3+m_2)}$

This problem is straightforward.

Problem 15) prove $\nabla(\log|\vec{r}|) = \frac{\vec{r}}{r^2}$

$\nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial x}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$ in Cartesian Coordinates.

$$\nabla(\log(r)) = \nabla\left(\log\left(\sqrt{x^2+y^2+z^2}\right)\right)$$

$$\frac{\partial}{\partial x}\log\left(\sqrt{x^2+y^2+z^2}\right) = \frac{1}{r}\frac{d}{dx}\sqrt{x^2+y^2+z^2} = \frac{1}{r}\left(\frac{2x}{2r}\right) = \frac{x}{r^2}$$

$$\Rightarrow \nabla(\log(r)) = \frac{x}{r^2}\hat{i} + \frac{y}{r^2}\hat{j} + \frac{z}{r^2}\hat{k} = \boxed{\frac{\vec{r}}{r^2}}$$

problem 14) $\nabla^2 f = \nabla\cdot\nabla f$

$$= \frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r} + \frac{1}{r^2\sin\theta}\left(\cos\theta\frac{\partial f}{\partial\theta} + \sin\theta\frac{\partial^2 f}{\partial\theta^2}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\varphi^2}$$

$$\Rightarrow \nabla^2\log(|\vec{r}|) = \frac{\partial}{\partial r}\left(\frac{1}{r}\right) + \frac{2}{r}\,\frac{1}{r} = -\frac{1}{r^2} + \frac{2}{r^2} = \boxed{\frac{1}{r^2}}$$

Problem 15) $F_d \propto bv^n$; m; v

a) time to stop? $[t] = T = [m]^\alpha [v]^\beta [b]^\gamma$; $[b] = \frac{ML}{T^2}\left(\frac{T}{L}\right)^n$

$$\Rightarrow T = M^\alpha\left(\frac{L}{T}\right)^\beta \frac{M^\gamma L^\gamma}{T^{2\gamma}}\,\frac{T^{\gamma n}}{L^{\gamma n}}$$

$$\begin{cases} M: \ M^0 = M^{\alpha+\gamma} \Rightarrow \alpha + \gamma = 0 \\ L: \ L^0 = L^{\beta-\gamma n+\gamma} \Rightarrow \beta = \gamma(n-1) \\ T: \ T^1 = T^{\gamma n-\beta-2\gamma} \Rightarrow \beta = \gamma(n-2) - 1 \end{cases}$$

$0 = \gamma + 1 \Rightarrow \boxed{\gamma = -1}$

$\Rightarrow \alpha = 1$; $\beta = 1 - n$

$\Rightarrow t \propto m$; $t \propto v^{1-n}$; $t \propto \frac{1}{b}$